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COMPLETE SPECIFICATION
(See Section 10 and Rule 13)

"MULTI-FUNCTIONAL BOREWELL RESCUE FRAMEWORK: IMPROVING SAFETY WITH STABLE LIFTING SYSTEM"

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The following specification particularly describes the invention and the manner in which it is to be performed:

FIELD OF THE INVENTION

The present invention relates to the fields of sensor technology, robotics, mechanical engineering, and real-time monitoring systems. It specifically pertains to an integrated framework designed to facilitate the safe and efficient rescue of individuals trapped in borewells, a significant safety concern in regions dependent on such water sources. By employing advanced sensing capabilities, a foldable mechanical support structure, and a robotic manipulation system, the invention addresses the multifaceted challenges of borewell rescue operations, aiming to enhance both safety and operational efficacy.

BACKGROUND OF THE INVENTION

Borewells serve as critical water sources in areas where surface water is scarce, supporting essential activities such as drinking, household use, agriculture, and industrial processes. In India, states like Tamil Nadu, Gujarat, Madhya Pradesh, and Haryana heavily depend on borewells due to their geographical and climatic conditions. These structures, typically ranging from 4 to 6 inches in diameter and extending 300 to 700 feet in depth, are vital for sustaining communities, particularly in rural and agricultural regions. However, the widespread use of borewells has led to a significant safety concern: accidents involving individuals, especially children, falling into open or abandoned borewells. Such incidents are distressingly common, posing severe risks due to the confined space, extreme depth, and hazardous environmental conditions within these structures.

The narrow diameter of borewells severely restricts access, making traditional rescue methods, such as manual extraction or digging parallel pits, both time-consuming and often ineffective. Compounding the challenge is the presence of harmful gases, including methane, carbon monoxide, and liquefied petroleum gas (LPG), which can accumulate within borewells. These gases create risks of suffocation, poisoning, or even explosions, endangering both the trapped individual and the rescue team. The lack of real-time data on the borewell's internal environment further complicates rescue efforts, as teams often operate

without precise information about gas levels, oxygen availability, or the victim's exact location. Existing rescue systems typically lack the integration of environmental monitoring, mechanical stabilization, and precise extraction mechanisms, leading to delays that can result in tragic outcomes. Consequently, there is an urgent need for an innovative, comprehensive solution that can address these multifaceted challenges, providing a reliable and efficient means of rescuing individuals from borewells while prioritizing their safety and survival.

OBJECTS OF THE INVENTION

The primary object of the present invention is to provide a compact and integrated borewell rescue system that enables the safe and efficient extraction of individuals trapped in narrow and deep borewells.

Another object of the invention is to monitor the borewell environment in real-time using sensors to detect harmful gases, oxygen levels, temperature, humidity, and to determine the victim's location.

A further object of the invention is to provide a deployable support mechanism that stabilizes the victim and prevents further descent during the rescue process.

Another object is to enable safe and precise retrieval of the victim using a remotely controlled robotic hand, operable through a sensor-equipped glove.

Yet another object is to enhance rescue coordination through wireless data transmission, cloud-based monitoring, and direct communication with the victim.

It is also an object of the invention to reduce physical risk to rescue personnel by automating key rescue functions and allowing remote operation.

A final object is to ensure affordability, portability, and ease of deployment using cost-effective materials such as 3D-printed structures and open-source electronics.

SUMMARY OF THE INVENTION

The invention provides a multi-functional borewell rescue framework that overcomes the limitations of conventional rescue methods by integrating advanced technological components into a cohesive system. The primary objective is to enhance the safety and efficiency of borewell rescue operations through real-time environmental monitoring, stable mechanical support, and precise robotic extraction. The system comprises three core subsystems: an air quality monitoring system, a support mechanism, and a robotic hand, each designed to address specific aspects of the rescue process. The air quality monitoring system employs sensors to detect harmful gases, measure oxygen levels, monitor temperature and humidity, and locate the victim, providing critical data to the rescue team. If oxygen levels are low, the system delivers oxygen through narrow tubes. The support mechanism, a foldable structure, creates a stable platform beneath the victim to prevent further descent and facilitate extraction. The robotic hand, controlled via a glove with flex sensors, mimics human hand movements to gently grip and retrieve the victim. These subsystems are interconnected through wireless communication modules and a central control unit, ensuring coordinated operation based on real-time data. By addressing environmental hazards, victim stabilization, and safe extraction, the invention significantly improves the likelihood of successful rescues, offering a practical solution to a critical societal problem.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings described herein are intended to provide a further understanding of the invention and are intended to be a part of the invention. However, the drawings as shown are representative for illustration and are non-limiting the scope of the invention. The drawings are described as below:

Figure 1(a) depicts the support mechanism in its insertion mode, showing a compact configuration for entry into the borewell.

Figure 1(b) illustrates the support mechanism in the deployment positioning stage, with the rectangular plate rotated for positioning.

Figure 1(c) shows the support mechanism in its base formation stage, with flaps unfolded to form a stable platform.

Figure 2(a) presents a schematic diagram of the experimental setup for the support mechanism, detailing its electronic components.

Figure 2(b) provides a schematic diagram of the experimental setup for the robotic arm, highlighting its control components.

Figure 2(c) depicts the schematic diagram of the air quality monitoring system's experimental setup, showing sensor integration.

Figure 3 illustrates the physical arrangement of air quality monitoring sensors at the top and at 3 feet within the borewell.

Figure 4(a) shows the air quality monitoring system positioned at the top of the borewell, while **Figure 4(b)** depicts it at a 3-foot depth. **Figure 4(c)** presents the entire air quality monitoring setup at the top, and Figure 4(d) shows it at 3 feet.

Figure 5 displays cloud-based data visualization of methane, CO, and LPG levels on the ThingSpeak platform.

Figure 6 shows real-time gas level measurements displayed on an LCD screen.

Figure 7 illustrates the 3D-printed hexagonal base support in open and closed positions.

Figure 8 depicts the experimental setup of the robotic hand system, including the control glove.

Figure 9 demonstrates the robotic hand grasping a toy, showcasing its gripping capability.

DETAILED DESCRIPTION OF THE INVENTION

Preferred features of the present invention will now be described with particular reference to the accompanying drawings. However, it is to be understood that the features illustrated in and described with reference to the drawings are not to be construed as limiting the scope of the invention.

The multi-functional borewell rescue framework represents a significant advancement in emergency rescue technology, specifically engineered to address the unique challenges of extracting individuals from borewells. The system operates in the confined and hazardous conditions of borewells, integrating advanced sensors, mechanical structures, and robotic components to ensure the safety of the victim and the efficiency of the rescue operation. The invention comprises three primary subsystems: the air quality monitoring system (200), the support mechanism, and the robotic hand, each meticulously designed to perform specific functions. Their seamless integration enables a coordinated and effective rescue process. The following sections provide an in-depth exploration of each subsystem, their components, operational mechanisms, and integration, with detailed references to the accompanying figures, culminating in a discussion of testing and validation outcomes.

Overview of the System

The borewell rescue framework is designed to address the critical challenges of rescuing individuals trapped in narrow borewells, typically 4 to 6 inches in diameter and up to 700 feet deep. The system integrates three subsystems to monitor environmental conditions, stabilize the victim, and facilitate safe

extraction. The air quality monitoring system assesses the presence of harmful gases, oxygen levels, temperature, humidity, and the victim's location, providing real-time data to guide the rescue. The support mechanism deploys a foldable platform to prevent the victim from falling further, while the robotic hand, controlled remotely, gently grips and retrieves the victim. These subsystems are coordinated through a central control unit, ensuring a cohesive operation. The system's design prioritizes safety, efficiency, and adaptability to various borewell conditions, making it a versatile solution for emergency rescues.

Air Quality Monitoring System (200)

The air quality monitoring system (200) is the foundation of the rescue framework, providing critical data on the borewell's environmental conditions. As illustrated in **Figure 2(c)**, the system (200) includes a NodeMCU microcontroller (202), which serves as the master controller, processing data from multiple sensors. The MQ-6 sensor (204) detects liquefied petroleum gas (LPG) in parts per million (ppm), operating by changing electrical resistance in the presence of LPG, with a detection range of 200 to 10,000 ppm. This sensor is crucial for identifying flammable gases that could cause explosions. The MQ-7 sensor (206) measures carbon monoxide (CO) levels in ppm, using a tin dioxide semiconductor that increases conductivity when exposed to CO, ensuring detection of this odorless, toxic gas. The MQ-135 sensor (208) monitors a range of hazardous gases, including methane, ammonia, and benzene, providing a comprehensive air quality assessment. The ME2-O2 sensor (210) measures oxygen concentration, typically in percentage, using an electrochemical process that generates a current proportional to oxygen levels, critical for ensuring a breathable environment. The DHT11 sensor (212) measures temperature in degrees Celsius and humidity in percentage, providing additional environmental data that affects both the victim's condition and system performance. An ultrasonic sensor (214) measures the distance to the victim in meters, using ultrasonic pulses to determine proximity, enhancing rescue precision.

Figure 3 illustrates the physical arrangement of these sensors, with the MQ-6 (204), MQ-7 (206), MQ-135 (208), and DHT11 (212) sensors positioned at the top of the borewell and at a depth of 3 feet to monitor conditions at different levels. An endoscopy camera (216), equipped with LED lights, provides high-resolution visual feedback of the borewell's interior and the victim's position, as shown in **Figure 4(a)** for the top position and **Figure 4(b)** for the 3-foot depth. **Figure 4(c)** depicts the entire air quality monitoring setup at the top, including the NodeMCU (202) and sensors, while **Figure 4(d)** shows the setup at 3 feet, highlighting the consistent sensor placement. The camera's visuals are essential for guiding the deployment of other subsystems.

The NodeMCU (202) processes sensor data and transmits it wirelessly using NRF24L01 modules (218), selected for their reliability in short-range communication. The data is also uploaded to the ThingSpeak IoT platform, as shown in **Figure 5**, which displays real-time graphs of methane, CO, and LPG levels, enabling remote monitoring by experts. **Figure 6** illustrates an LCD display (220) that provides on-site readings of gas levels, temperature, humidity, and distance, allowing immediate response to changing conditions. Walkie-talkie modules (222) facilitate communication with the victim, offering reassurance or instructions if the victim is conscious. The air quality monitoring system's comprehensive data collection and communication capabilities ensure that all rescue actions are informed by accurate and timely information, significantly enhancing safety and efficiency.

Support Mechanism

The support mechanism is designed to provide a stable platform within the borewell, preventing the victim from falling further and creating a secure base for extraction. As shown in **Figure 1(a)**, in the insertion mode, the support mechanism (100) consists of a mounting pipe (102), 13 inches long, which serves as the central shaft. Attached to its lower end is a rectangular plate (104), measuring 4 inches in length and 2 inches in width, with two trapezium flaps

(106) and (108), each 4 inches at the base and 1 inch in height, sloped at 45 degrees. In this configuration, the plate (104) and flaps (106), (108) are aligned parallel to the pipe (102), ensuring a compact profile for smooth insertion into the borewell's narrow diameter.

Figure 1(b) illustrates the deployment positioning stage, where the rectangular plate (104) rotates 90 degrees to become perpendicular to the mounting pipe (102), while the flaps (106) and (108) remain closed. This positioning prepares the mechanism to settle below the victim, guided by data from the ultrasonic sensor (214). **Figure 1(c)** depicts the base formation stage, where the trapezium flaps (106) and (108) unfold outward, controlled by servo motors (110), to form an extended, angled support platform. This configuration creates a broad, stable base capable of supporting the victim's weight and resisting the borewell's internal pressures.

The experimental setup of the support mechanism is shown in **Figure 2(a)**, including an Arduino UNO microcontroller (300), which processes data and controls the mechanism. An ultrasonic sensor (302) measures the distance to the victim, triggering deployment at the correct depth. NRF24L01 modules (304) enable wireless communication, while servo motors (306) control the unfolding of the plate (104) and flaps (106), (108). An LED indicator (308) signals when a safe deployment distance is reached. **Figure 7** illustrates the 3D-printed hexagonal base support, constructed using Hyper PLA, a cost-effective and durable material with a printing temperature of 190-230°C. The figure shows the mechanism in both open and closed positions, demonstrating its retractable design.

The support mechanism's design addresses the spatial constraints of the borewell, with its lightweight construction facilitating easy handling and its robust materials ensuring reliability. The ability to transition from a compact, folded state to a stable, expanded platform is a key innovation, providing a secure foundation for the victim during extraction.

Robotic Hand

The robotic hand is engineered to perform the delicate task of gripping and retrieving the victim, ensuring minimal risk of injury. As shown in **Figure 8**, the robotic hand system (400) includes a glove (402) equipped with flex sensors (404) on each finger, which detect the operator's hand movements by changing resistance when bent. These movements are processed by an Arduino UNO microcontroller (406), which transmits the data wirelessly via NRF24L01 modules (408) to the robotic hand. Servo motors (410), connected to the hand's articulated fingers, replicate the operator's movements in real-time, enabling precise control.

Figure 9 demonstrates the robotic hand (412) grasping a toy, showcasing its ability to securely hold objects, which is critical for safely retrieving the victim. The hand is constructed from lightweight, durable materials to operate effectively within the borewell's confined space. It works in coordination with the support mechanism, using visual feedback from the endoscopy camera (216) to position itself accurately. The control system prioritizes smooth and gradual movements to prevent sudden jerks, ensuring the victim's safety during extraction.

System Integration and Operation

The three subsystems are integrated through a central control unit that coordinates their operations based on real-time data, as programmed using the Arduino Integrated Development Environment (IDE). The NodeMCU (202) and Arduino UNO (300, 406) microcontrollers serve as the primary processing units, managing sensors, actuators, and communication modules. The rescue operation follows a structured sequence to maximize efficiency and safety.

The process begins with the deployment of the air quality monitoring system, as shown in **Figure 4(a)** and **4(b)**, which is lowered into the borewell to assess environmental conditions and locate the victim using the ultrasonic sensor (214)

and camera (216). The sensors (204, 206, 208, 210, 212) collect data on gas levels, oxygen, temperature, and humidity, which is displayed on the LCD (220) (**Figure 6**) and uploaded to ThingSpeak (**Figure 5**). If oxygen levels are low, oxygen is supplied through narrow tubes. Based on the victim's location, the support mechanism (100) is lowered in its folded state (**Figure 1(a)**), positioned below the victim, and deployed by rotating the plate (104) (**Figure 1(b)**) and unfolding the flaps (106, 108) (**Figure 1(c)**). The robotic hand (412), controlled via the glove (402) (**Figure 8**), is then lowered and positioned using camera visuals, gently gripping the victim (**Figure 9**). With the victim secured, the support mechanism and robotic hand are retracted, bringing the victim to the surface.

Continuous monitoring and communication are maintained throughout, with the LCD (220) providing on-site data and ThingSpeak enabling remote oversight. Walkie-talkie modules (222) facilitate communication with the victim, offering psychological support. The system's integrated approach ensures that each stage is informed by accurate data and executed with precision, minimizing risks and maximizing success.

Testing and Validation

The borewell rescue framework was fabricated, tested, and operated at Chaitanya Bharathi Institute of Technology, Hyderabad, in a mock borewell setup simulating real-world conditions. The air quality monitoring system demonstrated high accuracy in detecting gas levels, oxygen, temperature, and humidity, with sensor readings aligning with expected values, as visualized in **Figure 5** and displayed in **Figure 6**. The support mechanism successfully transitioned through its stages (**Figures 1(a)-(c)**), forming a stable platform, as shown in **Figure 7**. The robotic hand performed precise gripping tasks, as demonstrated in **Figure 9**. The experimental setups for each subsystem, depicted in **Figures 2(a)**, **2(b)**, and **2(c)**, confirmed their reliability and integration. The trials validated the system's ability

to address environmental hazards, stabilize the victim, and execute safe extraction, highlighting its potential to improve borewell rescue outcomes.

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We claim:

1. A multi-functional borewell rescue system comprising:
    a. an air quality monitoring subsystem configured to detect hazardous gases including methane, carbon monoxide, and LPG, measure oxygen concentration, temperature, and humidity, and determine a victim's location using sensors and an ultrasonic module;
    b. a support mechanism comprising a foldable platform configured to be inserted into a borewell in a compact state, deployable to form a stable base below the victim, said mechanism actuated by servo motors;
    c. a robotic hand subsystem comprising a robotic arm with articulated fingers controlled by a glove equipped with flex sensors for mimicking human finger movements;
    d. a control system comprising at least one microcontroller configured to process sensor data and actuate the support mechanism and robotic hand;
    e. a wireless communication module for transmitting real-time data from the sensors and control signals to and from the system;
    f. and wherein the system is configured to transmit environmental and victim location data to a cloud-based platform for remote monitoring and expert consultation.

2. The system as claimed in claim 1, wherein the air quality monitoring subsystem includes an MQ-6 sensor for LPG detection, an MQ-7 sensor for CO detection, an MQ-135 sensor for general air quality, a ME2-O2 sensor for oxygen levels, and a DHT11 sensor for temperature and humidity.

3. The system as claimed in claim 1, wherein the air quality monitoring subsystem includes an endoscopy camera with LEDs for real-time visual feedback inside the borewell.

4. The system as claimed in claim 1, wherein the support mechanism includes a rectangular plate and trapezium flaps that rotate and unfold respectively, to form a stable support base upon receiving a trigger signal based on the victim's detected depth.

5. The system as claimed in claim 4, wherein the support mechanism is constructed using 3D-printed Hyper PLA material for lightweight, strength, and cost-effectiveness.

6. The system as claimed in claim 1, wherein the robotic hand subsystem is operable via a glove comprising flex sensors, wherein movement of human fingers is translated into proportional motion in the robotic fingers through servo motors.

7. The system as claimed in claim 1, wherein the communication between the glove and the robotic hand is enabled through NRF24L01 wireless transceivers.

8. The system as claimed in claim 1, wherein the control system includes at least one NodeMCU and Arduino UNO microcontroller for data acquisition, processing, and actuation.

9. The system as claimed in claim 1, wherein the wireless communication module is configured to upload sensor data to a cloud-based IoT platform such as ThingSpeak for visualization and expert remote access.

10. The system as claimed in claim 1, wherein a walkie-talkie module is provided to facilitate real-time audio communication between rescuers and the victim.

11. The system as claimed in claim 1, wherein oxygen can be supplied through a narrow tube into the borewell when low oxygen concentration is detected.

12. The system as claimed in claim 1, wherein the support mechanism is equipped with an LED indicator to signal the appropriate deployment point.

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ABSTRACT

MULTI-FUNCTIONAL BOREWELL RESCUE FRAMEWORK: IMPROVING SAFETY WITH STABLE LIFTING SYSTEM

The invention relates to a multi-functional borewell rescue system designed for the safe retrieval of individuals trapped in narrow borewells. The system integrates an air quality monitoring unit equipped with sensors to detect hazardous gases, oxygen levels, temperature, and humidity, along with an ultrasonic sensor and endoscopy camera to locate the victim. A deployable mechanical support mechanism, made of 3D-printed material, provides a stable base below the victim. A robotic hand, remotely controlled via a glove with flex sensors, enables gentle and precise extraction. Wireless modules ensure real-time communication between components and allow for remote monitoring through cloud platforms. The system enhances rescue efficiency, minimizes risks to both the victim and rescue personnel, and offers an affordable and portable solution adaptable to various borewell scenarios.